

Data Gathering

CMPSC 105 – Data Exploration



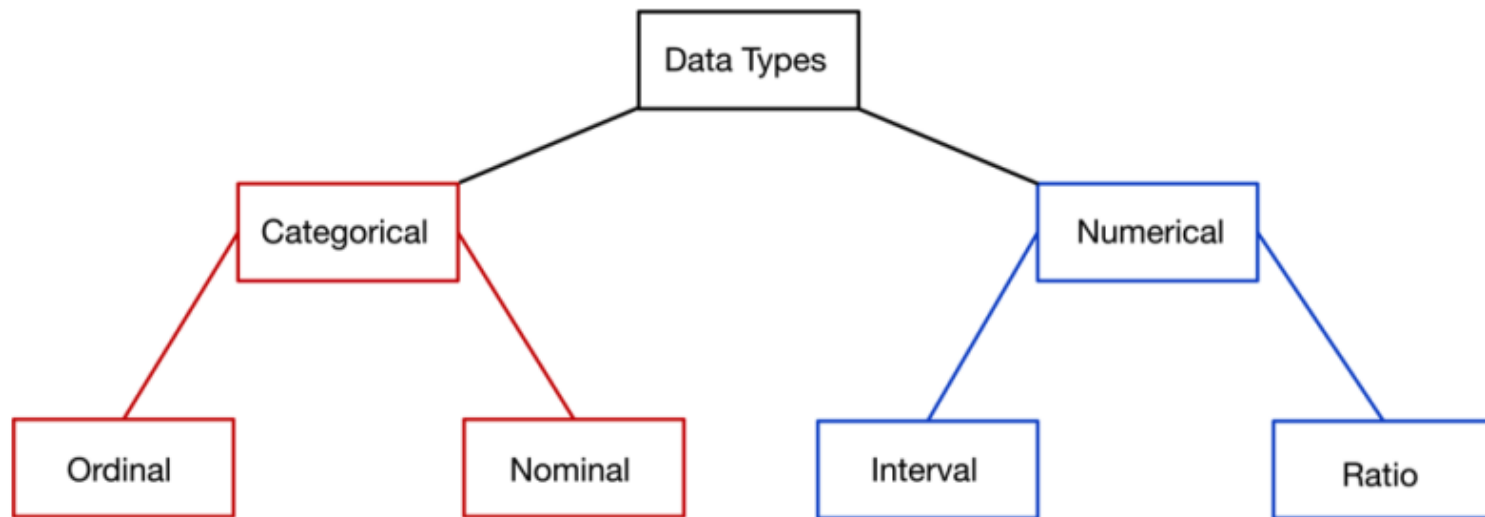
ALLEGHENY COLLEGE

Goals

- What **types** of data can you gather?
- What **sources** can data come from?
- How is data **stored**?

Data Types

Reading: [Types of Variables](#)



and numerical discrete!

Categorical Nominal

- Name only, qualitative
 - car makes
 - zip code
 - eye color
- Dichotomous variable refers to a categorical nominal variables with only two values
 - left handed / right handed
 - current student / non student

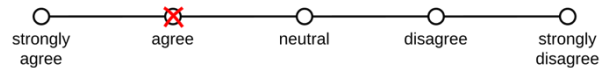
Categorical Ordinal

- ordered, quasi-quantitative
 - agreement level
 - education level
- ordinal data can seem almost quantitative, but there is no way to know the value of the levels
- ordinal scales are usually used to measure non-numeric features like happiness, customer satisfaction
- Likert Items are often used to obtain categorical ordinal data

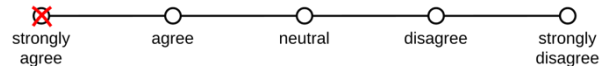
Likert Items

Website User Survey

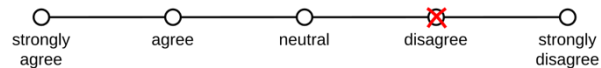
1. The website has a user friendly interface.



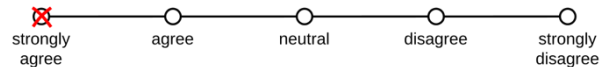
2. The website is easy to navigate.



3. The website's pages generally have good images.



4. The website allows users to upload pictures easily.



5. The website has a pleasing color scheme.



Numerical Data

- Numerical data are quantitative
- Discrete means countable
 - number of students
 - number of views or likes

Discrete and Countable Objects



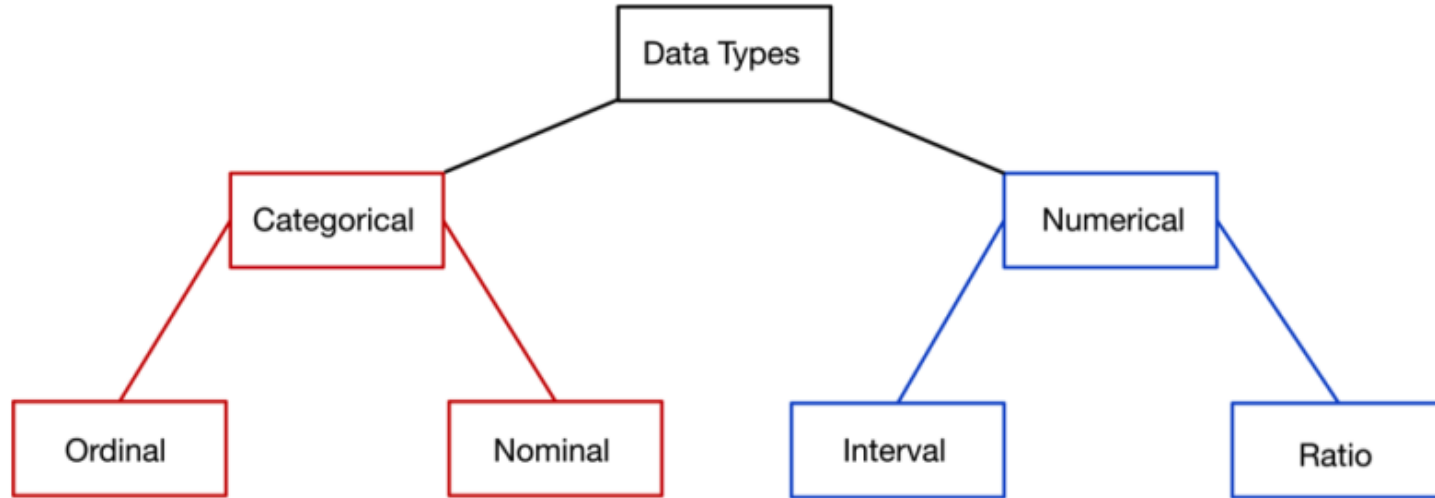
- *Discrete* means “countable” (can be listed in an order)
- We can count the number of animals.

Numerical Data

- Numerical data are quantitative
- Continuous means measured
 - height must be measured
 - Interval refers to continuous data with no reference point
 - Temperature, we know 28 F vs 33 F is a difference of 5, but does a difference of 5 mean the same thing between 100 and 105?
 - Interval data cannot be divided meaningfully. Don't compute what is 1/5 of the temperature today
 - Stats use division, causing a problem for interval data
 - Ratio refers to continuous data with a reference point
 - Height, weight, distance, speed
 - division or other ratios are meaningful.

Practice

Recall an example for each data type

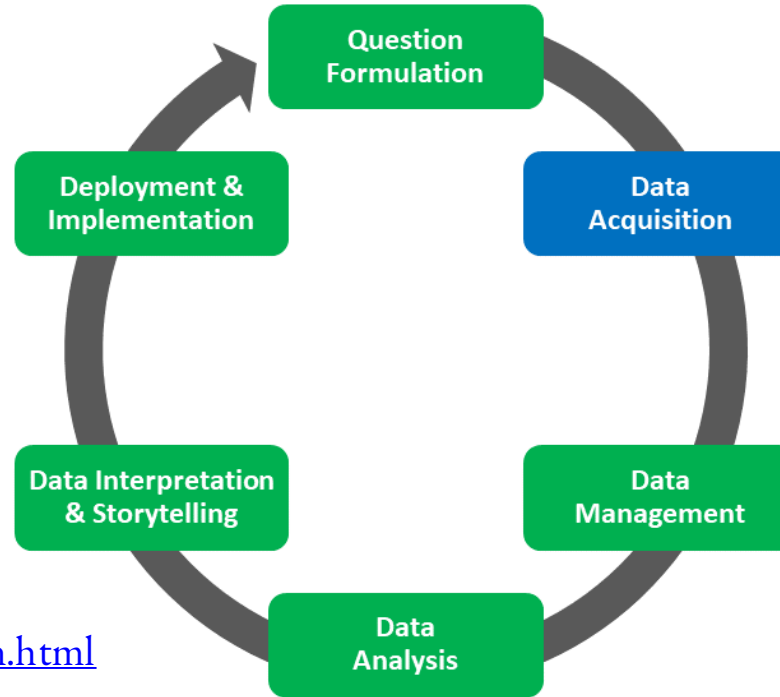


and numerical discrete!

[Data Gathering Form](#)

Data Sources

- Pre-existing data
- Extracted data
- Acquired data



<https://rforhr.com/dataacquisition.html>

Data Acquisition

- Getting brand new data using surveys and sensors!

Surveys for Acquisition

- Surveys are directed toward people
- Surveys answer subjective perceptual and behavioral questions, primarily
- Different questions produce different data types
 - multiple choice questions
 - free response questions
 - slider ratings
 - Likert scales

Surveys for Acquisition

- Survey development should be guided by a hypothesis
- Only questions that have appropriate data types for analysis should be used
- Quality controls should be in place (e.g. timestamps, inclusion criteria)
- Data security protocols should be in place to protect private data
- Human Subjects Research must be reviewed by the Institutional Review Board
- Reading: [Surveys](#)

Sensors for Acquisition

- Sensors can measure inanimate objects/materials or people/animals
- Electronic sensors for things
 - motion detection
 - freezer thermometer
 - tire pressure sensor
- sensors for people
 - motion capture
 - body temperature
 - electro-physiological
 - ECG - electrocardiogram
 - EEG - electroencephalogram
 - electrodermal

Sensors for Acquisition

youtube ([sensors](#))

Acquisition Drawbacks

- Time consuming
- Data management is required (persistent storage, security)
- Maybe the data already exist somewhere
- Maybe surveys and sensors cannot answer the question you have

Extracted Data

Extracting Data from Documents

- Documents often contain information but it is not in an analyzable format
 - Author names on research papers
 - Foreground of an image
 - Background of an image
 - Video transcripts of speech
 - Video transcripts of sounds
 - Cells (biology) appearing in images
 - Language structure
 - noun, verb, adj, conjunction,
 - word length
 - num vowels
 - num syllables
 - etc

Extracting Data by Web Scraping

- Scraping is gathering data from websites by reading into the html structure to get the content out of specific html tagged elements.
- Scraping tools exist to assist with formatting
 - Requests, BeautifulSoup, Selenium
 - example with https://en.wikipedia.org/wiki/Tadasu_no_Mori

```
import requests
# Making a GET request
r = requests.get('https://en.wikipedia.org/wiki/Tadasu_no_Mori')
# print content of request
print(r.content)
```

Extracting Data by Web Scraping

```
import requests

url = "https://en.wikipedia.org/wiki/Tadasu_no_Mori"

headers = {
    "User-Agent": "CMPSC105 Data Gathering Demo"
}

r = requests.get(url, headers=headers)

print(r.status_code)
print(r.text[:1000]) # print first 1000 characters
```

Extracting Data & Storage

- Extracted data must be entered into usable structures
 - databases
 - spreadsheets
 - plain text files

Drawbacks of Extracting Data

- data entry is time consuming
- data should be consistent, but real data is never perfect, so compromises must be made
- human errors can be difficult to detect
- data entry done with automated entry tools is hard to validate
- validation takes human time, or accepted inaccuracy
- arbitrary decisions may have to be made

Pre-existing Data

Sources of Pre-existing Data

- Tableau Data
- Others

Drawbacks of Pre-existing Data

- Ethical collection practices may or may not have been used
- data have been used repeatedly in explorations leading to bias and overfitting on the data across explorations
- maybe the pre-existing data cannot answer the question you have

Data Fraud

- [Image fraud Article PDF](#)
- <https://www.youtube.com/watch?v=GeAPtIdb4Qs>

CSV

- Comma-separated values format
- plain text file
 - readable and writable in many software programs
 - Google Sheets
 - MS excel
 - Notepad
 - Sublime Text
 - VS Code
- Reading CSV [Section 2.4 & 2.4.1](#)

CSV

Each column separated
by a comma

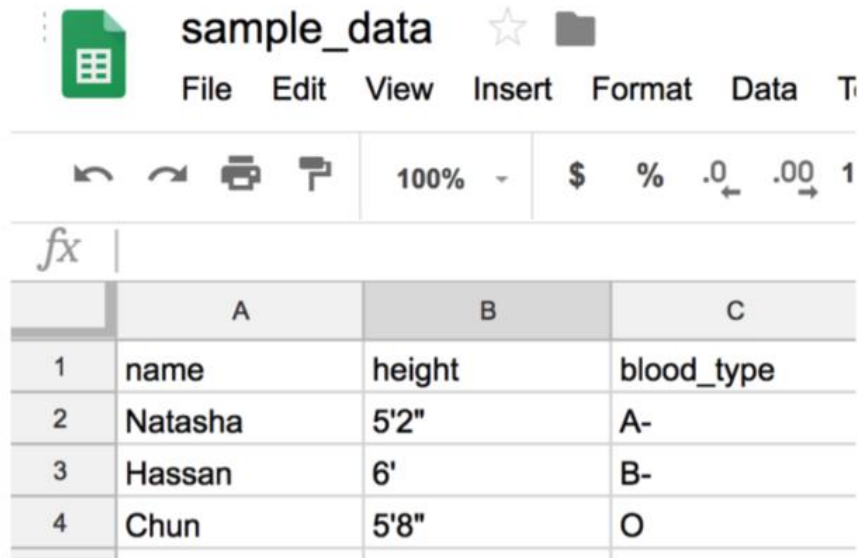
CSVs

sample_data.csv — Edited

```
name, height, blood_type
Natasha, 5'2", A-
Hassan, 6', B-
Chun, 5'8", O
```

Has the
extension
".csv"

Each row is separated
by a new line



	A	B	C
1	name	height	blood_type
2	Natasha	5'2"	A-
3	Hassan	6'	B-
4	Chun	5'8"	O

CSV Data

- Suppose you had some data in a CSV format
- How to do something with the data?!

- CSV data: `sandbox/contacts.csv`

```
tylernelson@gmail.com,Careers adviser  
gregory02@medina-mayer.com,"Accountant, management"  
jonesmiguel@hotmail.com,Health and safety inspector  
rsanchez@yahoo.com,"Surveyor, planning and development"  
hillfrank@ward-wood.com,"Scientist, physiological"  
aaronhunter@gmail.com,"Surveyor, insurance"  
kylebarnes@hotmail.com,Records manager  
joe70@yahoo.com,Network engineer  
torresjames@white.info,Electrical engineer  
shawkins@watson.com,Science writer
```

Functions that Manage Tuples

File: csvreader.ipynb

```
from os.path import exists
from logging import exception

def openCSVFile(fname_str: str) -> str:
    """loads a file, returns csv string"""
    # print("openCSVFile()")
    if not exists(fname_str): # no file found?
        print(f"\t [-] No file by that name: {fname_str}")
        exit() # end program if no file has been found.

    try:
        data_str = open(fname_str, "r").read()
    except exception:
        print("\t [-] Using correct filename?")
        return None

    # commas in this loaded file?
    if len(data_str) > 0 and "," in data_str:
        return data_str

    return None
```

Functions that Manage Tuples - iterateData

```
def iterateData(in_str: str) -> dict:
    """Function to output the data in tidy lines. Place data into dictionary."""
    contact_dict = {}
    for line in in_str.splitlines():
        # print(line)
        # get the name, located before first comma
        name_str = line[: line.find(",")]
        service_str = line[line.find(",") + 1 :].replace("'", "")
        contact_dict[name_str] = service_str
    return contact_dict
```


Functions that Manage Tuples - main()

```
def main() -> None:
    """driver function"""
    prompt_str = "\t Enter the CSV filename : "
    myFile_str = input(prompt_str)
    # print(f"\t [+] You entered file : {myFile_str}")

    myCSV_str = openCSVFile(myFile_str)
    # print(f"Main() {myCSV_str}")

    # print out in tidy lines
    myContact_dict = iterateData(myCSV_str)

    # print(f"Dictionary of names: {myContact_dict}")
    for i in myContact_dict:
        print(f"\t [+] {i} : {myContact_dict[i]}")
```

CSV Activity

- Make a CSV file based on this one image to explore this open question: Are healthy foods too expensive?
- How many different things can you imaging recording off of this one image?!



CSV Activity Details

Start a Google Sheet. Include:

- Store Name
- Date
- Time
- Location
- Total number of items purchased
- Total number of non-food items purchased
- Total amount of “health” foods
- Total cost in USD
- Total cost of “health” foods
- Total cost of non-food items

CSV Activity Details

Submit your work to your Repo

Use the name “data-gathering-activity.csv”